

## Digital Logic Design

(EE1005)

Date: 12<sup>th</sup> April, 2025

Course Instructor(s)

Ms. Tamania Javaid

Mr. Ahmad Hamza

## Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 45

Total Questions: 3

Roll No

Section

Student Signature

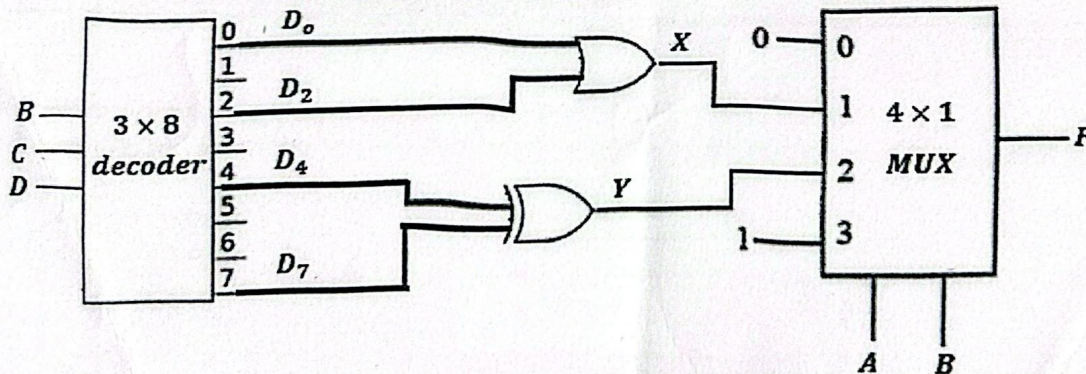
- Attempt all the questions.
- Show complete working of each question.
- Multiple solutions of the same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

**CLO #3: Design combinational circuits using functional blocks.**

Q1:

[15 marks]

Construct the truth table of the following digital system.



Take A, B, C, D as inputs and add columns to show X, Y and F as shown below. Make completely filled table on your answer sheet.

| A | B | C | D | X | Y | F |
|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |

**CLO #3: Design combinational circuits using functional blocks**

Q2:

[15 marks]

Design a combinational circuit that take a 3-bit number as input and generates its 1's complement as output. Implement the circuit using 3-8 line decoder with active low enable and active high output and minimum additional logic

**CLO #2: Design combinational circuits using functional blocks**

Q3:

[15marks]

Design a digital circuit that takes a 4 bit number X as input and increments it by 2 at the output. Use 1 bit full adder as a functional block. You can add the extra logic if required with any two input gates. Show complete working.